

User Evaluation of Group-in-a-box Variants

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ABSTRACT

Group-in-a-box (GIB) is a graph-drawing method designed to facilitate the visualization of the group structure of a graph. GIB allows the user to simultaneously view group sizes and inter- and intra-group structures. Several GIB variants have been proposed in the literature; however, their advantages and disadvantages have not been studied from the perspective of human cognition. Therefore, herein, we used eye tracking analysis and user surveys to evaluate the user experience of four GIB variants: Squarified-Treemap GIB (ST-GIB), Croissant-and-Doughnut GIB (CD-GIB), Force-Directed GIB (FD-GIB), and Tree-Reordered GIB (TR-GIB). We found some trade-offs among the methods for each type of user task and that FD-GIB and TR-GIB are superior than the other variants. Although ST-GIB's results were good, links were difficult to read in this graph layout. Eye-tracking data was gathered to determine which elements in each visualization significantly affected user experience. The results of this study will promote the effective use of GIB to analyze networks such as social networks or web graphs.

Index Terms: Human-centered computing—Visualization—Visualization techniques—Graph drawings; Human-centered computing—Visualization—Visualization design and evaluation methods

1 INTRODUCTION

In the field of graph drawing, numerous methods have been proposed for visualizing network data [14, 38, 40], with each method aiming at a different visualization goal. Some methods are designed for the analysis of graphs representing real-world relationships such as social networks and web graphs. These networks are often characterized as complex [29] because they tend to have community structure [15, 30]. A network with community structure contains groups of nodes which have dense within-group connections and sparse between-group connections. For example, networks of Twitter accounts represent users as nodes; pairs of users that mention each other are connected with edges, which can be weighted according to the number of mentions. In such a network, users can be divided into groups using the topology or other features of the graph. When a network is larger or has higher density, observers may face more difficulty in understanding a graph of the network because the high numbers of nodes, edges, and groups may be cluttered. One general method for distinguishing groups in a graph is to present nodes with different colors or shapes; however, this method is not effective for a large or dense network. To analyze real-world data, effective visualization techniques for understanding the group structure of a network is essential.

Group-in-a-box (GIB), illustrated in Figure 1, is a graph-drawing method designed for networks having a community structure.

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Among the many available community-structured methods for visualizing graphs, GIB is well-studied and has several variants with different visualization goals [8, 32, 37]. GIB arranges all nodes in a group into a box that has an area proportional to the number of nodes in the group such that each group can be visualized separately. The GIB layout enables simultaneous visualization of group structures, group relationships, inter-group relationships, and the sizes of the groups in a graph. Although some computational experiments [8, 32] have evaluated GIB variants, to the best of our knowledge, no user experiments have tested variations of the proposed approach. Some network measures were used in the abovementioned computational experiments (e.g., the number of edge crossings) that are known to affect the readability of a graph [3, 34–36]. Graph readability is also affected by other more-subjective factors, which are easiest to evaluate through user testing.

Therefore, we employed user testing to ascertain which GIB variant offers the most utility to users. Study participants tested the Squarified-Treemap GIB (ST-GIB), Croissant-and-Doughnut GIB (CD-GIB), Force-Directed GIB (FD-GIB), and Tree-Reordered GIB (TR-GIB) variants. ST-GIB uses the squarified treemap algorithm [5] to lay out the group boxes. CD-GIB arranges the boxes according to the number of other boxes connected to it, which may result in less visual clutter. A force-directed layout scheme is used to arrange the boxes in FD-GIB, whereas TR-GIB is produced by reordering boxes generated using the ST algorithm to minimize group proximity. We implement four types of tasks in user tests designed on the basis of survey studies by Vehlow et al. [40] and Saket et al. [38] to observe each layout's features from various perspectives. We measured the accuracy and completion time for each task.

We also collected eye-tracking data during the tests. Eye tracking is a technique used to record gaze data, and it is often used in the field of human-computer interaction. Gaze data is useful for understanding task performance in our tests because the gaze data of participants can grant insight into why a specific layout achieves better results than the others [1, 11, 25]. Specifically, we use gaze data to understand which elements in a layout help or hinder users when performing the experimental tasks. These elements may also have an influence on the readability of other visualization techniques.

One of our main contributions is the evaluation of four GIB layouts and the determination of the best GIB. We assume that each GIB has advantages and disadvantages. Thus, we aim to evaluate these GIB variants through four tasks and verify which GIB is significantly superior in each task. We also provide some evidence to support our results using the eye-tracking data to clarify which elements in a particular visualization affect the results. Finding these elements might contribute to the improvement of both GIB and other visualization techniques.

The four GIB layout schemes have also been made available to the public. Our website provides GIB images, as well as further analysis of the networks [2]. These schemes can help researchers in analyzing their network data and may even encourage new findings.

2 RELATED WORK

In this section, we review two types of related studies. The first type concerns graph drawing methods for networks having community structure, and the second type concerns the evaluation of visualization using eye-tracking data.

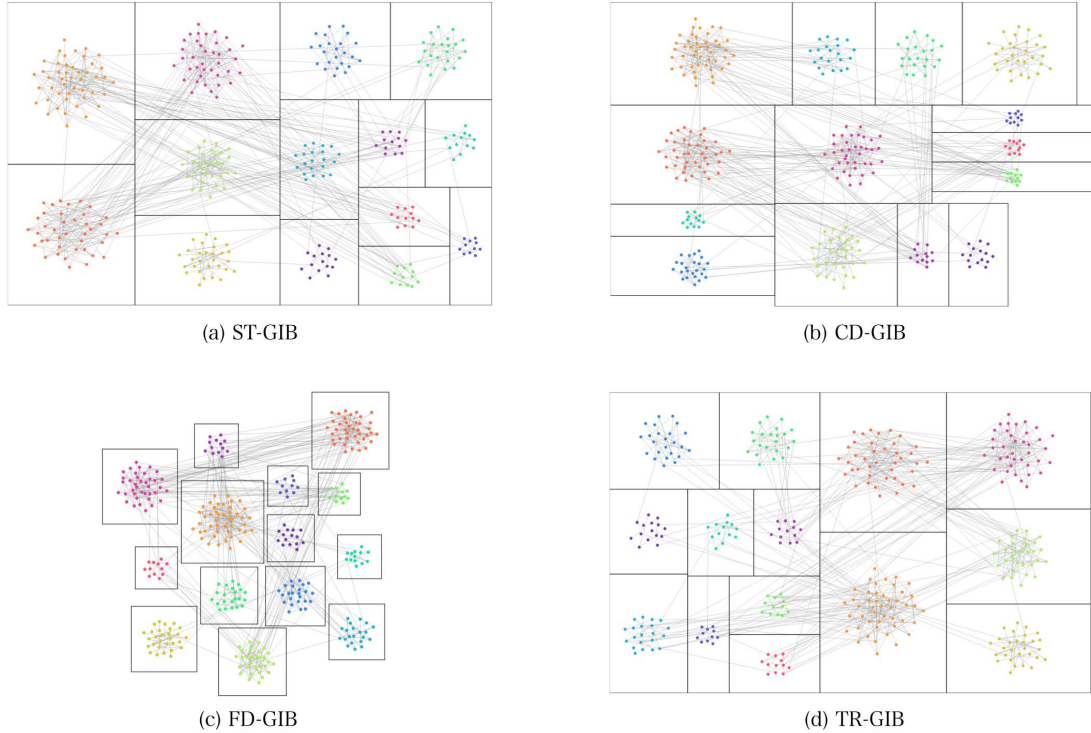


Figure 1: Examples of group-in-a-box (GIB) layouts. The four GIB layouts look different but the original data are the same. (a) ST-GIB is based on tree mapping. (b) CD-GIB considers the links connecting the groups. (c) FD-GIB arranges the boxes with a force-directed layout. (d) TR-GIB is generated by reordering the boxes generated in the ST-GIB layout to minimize group proximity.

2.1 Graph-drawing Method for Group Structure

Graph drawings are used to design a layout for complex networks. Eades proposed a method using force-directed placement in which attractive and repulsive forces are balanced to produce an optimal equilibrium [12]. This force-directed method has been widely used [23], and several approaches are also available for this purpose (see, e.g., [9, 16, 19, 24]).

However, the recent popularity of network visualization has led to the development of techniques specifically designed for visualizing networks with group structure. In large-scale network data, the importance of community detection is known and several approaches have been invented, often based on a force-directed layout. One typical method to illustrate the group structure of a graph is varying the color [26] or shape of nodes; but this method is often ineffective in real-world networks.

Vehlow et al. provided a survey of the visualization of group structures in graphs [40]. They described a taxonomy of visualization methods and tasks employed in user testing. This visualization taxonomy is classified as follows: vertex visualization and vertex group structure. The former represents visual grouping of nodes, either by node attributes, juxtaposition, superimposition, or embedding. The latter depends on whether a network can be characterized by group overlapping and/or the hierarchy of the graph structure. A node in a network with group overlapping may simultaneously belong to multiple groups. A hierarchical network features a hierarchical structure among groups.

The GIB layout is also a visualization for group structure; it is presented in [8, 32, 37]. This method is categorized as a superimposed visualization and is best for networks without group overlapping in Vehlow's taxonomy. ST-GIB and TR-GIB can be also applied for

networks with hierarchical structure. Chaturvedi et al. [8] compared three types of GIBs: ST-GIB, CD-GIB, and FD-GIB, by calculating several objective measures: edge-box overlap, percent screen space wasted, execution time, and mean group-box aspect ratio. They also provide several case studies.

Although such objective evaluation is effective, it is insufficient because certain hidden factors such as human cognition also influence readability. A user study in which participants use the graphs to complete multiple tasks could facilitate a more-thorough discussion of the layout's effectiveness and appropriately evaluate GIBs while considering such hidden factors. The algorithm proposed by Didimo and Montecchiani [10] also produces a good layout for visualizing group structure with arranging groups in boxes; however, this looks similar to FD-GIB. For the sake of easy comparisons, we focus on only four GIB layouts in this study.

2.2 Evaluation through Eye Tracking

Several studies have evaluated visualization techniques through eye tracking [7, 22, 27, 28, 33], which have been widely used to collect gaze data and measure the participants' visual attention [1, 11, 25]. Such systems are often used to analyze how well participants perform in a user study. Researchers can elicit clues about why one visualization is better than another by analyzing gaze data.

Burch et al. evaluated three variants of tree diagrams through eye tracking [7]. Their analysis of eye-tracking data explained why the task-completion results differed, as they found that the task with the layout with the least accuracy required users to look over complex cross-check trajectories.

Several studies provide guidelines on the visual analytics of eye-tracking data [1, 11, 25]. Eye-tracking datasets are often very large

and difficult to analyze; hence, spatiotemporal gaze data requires specific methodologies for analysis. Burch et al. mentioned two methods for analyzing gaze data [6], focusing on either areas of interest (AOIs) or gaze trajectories. Since early studies, researchers have known that although AOI plays a crucial role in the analysis process of eye movement data, people concentrate on interesting and informative regions of a display [41]. GIB divides the screen into boxes that are naturally regarded as AOIs. Thus, AOI analysis was also used in the present study as it is particularly in the analysis conducted herein.

3 GIB LAYOUTS

This section describes the four evaluated GIB variants, examples of which are shown in Figure 1. Each graph in this figure has a different GIB layout, but all layouts are representations of the same network data. As regards evaluation, Chaturvedi et al. have performed a computational experiment on three of these layouts: ST-GIB, CD-GIB, and FD-GIB [8]. Onoue et al. showed that TR-GIB is advantageous over ST-GIB in terms of computational measures [32]. Our target layouts are the four abovementioned layouts which need to be evaluated from the perspective of human cognition through user experiment.

3.1 ST-GIB

The squarified treemap GIB (ST-GIB) (Figure 1(a)) proposed by Rodrigues et al. [37] is based on the squarified treemap algorithm developed by Bruls et al. [5]. Bruls et al.'s method was originally designed for tree mapping, which is a visualization method that considers rectangular regions and numerical columns as inputs to divide a region into tiles, whose areas are proportional to their values [39].

In ST-GIB, each group is considered a vertex of the treemap and is depicted in the shape of a tile with nodes belonging to the group. The treemap algorithm facilitates a space-filling arrangement with boxes having low aspect ratios, which is important for analyzing the box's content [5]. However, the ST-GIB layout exclusively uses the squarified treemap algorithm to arrange groups; hence, the relationships among nodes are not considered when drawing the tiles. As a result, links frequently overlap, significantly hampering the user's understanding of the network [3, 34–36]. To arrange tiles using ST-GIB, we utilized the squarify Python library (<https://github.com/laserson/squarify>) that implements Bruls' algorithm. This method arranges the boxes in order of box size, which simplifies the comparison of group sizes for users.

3.2 CD-GIB

Chaturvedi et al. [8] proposed the croissant-and-doughnut GIB (CD-GIB) (Figure 1(b)). They developed this layout to improve ST-GIB such that it considers link information connecting a node to another node belonging to a different group. In particular, they arranged tiles based on the G-degree and G-skewness. A group's G-degree is defined as the number of other groups it is connected to, and its G-skewness is the proportion of nodes included in the two most-connected groups (groups with the highest G-degree). A layout is then chosen from croissant-GIB, doughnut-GIB, or ST-GIB according to the total number of groups and the G-skewness of the graph. Chaturvedi et al. defined criteria for appropriately using the three methods, and we applied the same criteria.

Croissant-GIB places the most-connected box, which has the highest G-degree, at the center-top of the screen. The other boxes are set such that they surround the most-connected box, resulting in a croissant shape. Another layout, called doughnut-GIB, assigns the most-connected box to the center with other boxes surrounding it to form a doughnut shape.

The most-connected group is placed close to the center to reduce edge overlaps; hence, the readability is better than that when using

ST-GIB. However, the aspect ratios of the boxes tend to be less uniform, which makes the area of the boxes difficult to estimate.

3.3 FD-GIB

Chaturvedi et al. [8] also proposed the FD-GIB layout (Figure 1(c)). In FD-GIB, the group boxes are arranged with a force-directed layout run on the whole network, with the vertices for this layout representing entire groups and the edges between them representing the links between groups. Then, group boxes are overlaid on this initial layout. FD-GIB can cause group overlaps, which are then removed using the PRISM method [13].

Chaturvedi et al. used the Harel–Koren fast multiscale layout [20] to arrange the rectangles in FD-GIB, but we use the D3.js force simulation [4] because of its easy availability and good results. However, as reported by Chaturvedi et al. [8], according to experimental evaluations of several layout algorithms performed by Hachul et al. [17, 18], several good options are available [20, 24].

This layout clearly displays the aggregate topology, but it uses more screen space. The area of each group must be smaller than that of the other layouts; hence, the relationships within a single group are more difficult to observe. However, this layout can maintain a consistent aspect ratio because of which participants can easily recognize the size differences between boxes.

3.4 TR-GIB

Onoue et al. developed TR-GIB to minimize the weighted sum of distances between groups by reordering the sibling nodes laid out by ST-GIB [32]. An example of this layout is shown in Figure 1(d). This method is visually similar to ST-GIB, but the tiles are reordered by solving an optimization problem to reduce the total length of the links between groups. As treemaps, such as ST-GIB, have a tree-like structure, vertices with the same depth in a tree can be reordered. The shorter edges in TR-GIB reduce the number of edge crossings, thereby improving the readability of the graph. We expect this layout to have the advantages of ST-GIB (i.e., good aspect ratio and screen efficiency) with an additional advantage of fewer edge overlaps.

4 DATA FOR EXPERIMENT

We describe the data used for our user tests in this section. The data for GIB is a network diagram comprising nodes, links, and groups. The network data used in the tests was analyzed using several computational measures before testing began.

4.1 Data and Layout Generation

We used random network data for these experiments, and each graph included multiple groups and some strong edge connections between certain group pairs similar to real-world data. This experimental data was generated as follows, considering the efficiency of generation, similarity to real data, and the difficulty of the tasks in the tests.

The networks were generated through a procedure similar to that used by Onoue et al. [32]. Our method differs from that of Onoue et al. in two key aspects. The first is how we randomly chose the number of groups from a normal distribution with m_{mean} and m_{stdev} ranging from m_{min} to m_{max} . Second, we determined the number of vertices in a vertex set V_i based on random numbers from the normal distribution with v_{mean} , v_{stdev} , and a minimum of v_{min} . We calibrated the parameters shown in Table 1 to have our random network resemble the actual Twitter data reported by Chaturvedi et al. [8]. Although we do not intend to focus on Twitter networks, that structure is a paradigm case for which GIB was designed—a complex network with group structure. The parameters in Table 1 were set through several processes. First, we calibrated the dataset to make the numbers of groups, nodes, and links correspond to those reported by Chaturvedi et al. However, this yielded more nodes and links than we could interpret; hence, we reduced the number of

nodes by multiplying v_{mean} and v_{stdev} by 0.4 and p_{in} , p_{bridge} , and p_{out} by 0.3. This process yielded the graphs shown in Figure 1.

After generating the data, we individually applied the four layouts to the data to obtain tile arrangements. The four layout schemes arrange only boxes; hence, we had to determine the layout of the nodes within each box. We used the standard force-directed layout, in particular D3.js force simulation [4]. This method is widely used in graph drawing [23]. The nodes are subjected to repulsive between-node forces, attractive forces between adjacent nodes, and gravity from the center of the group tile to which they belong. Although the layout inside the box affects the task results, the force layout is known to reduce edge crossing and is generally comprehensible; hence, we consider this layout suitable. The attractive force from other groups may move the nodes of a group to the outside of the box, which we avoided by laying the nodes within a group independent of other groups.

Edges in graph visualizations are often represented as straight or bundled lines. A graph drawing with only straight lines can become very complex, and edge bundling is effective when there are too many edges. However, straight lines directly represent the relationship between two nodes, and are easier to discriminate; hence, we represented edges with straight lines. These link-drawing and node-placement methods are different from the original GIB methods [8, 32, 37], but we suppose that each layout retains original characteristics of the network data such that the results of user testing will offer clear distinctions between the four box-layout schemes.

4.2 Computational Measures for Network Data

Various metrics are known that can affect the readability of graph drawing and provide clues about the results of the user tests. We computed four measures of the visualization before user testing, namely the number of edge crossings, the uniformity of edge lengths, screen space efficiency, and the mean aspect ratio of the group boxes. We chose these four metrics to primarily resemble those used by Chaturvedi et al. [8]; however, we used edge crossings rather than the overlap of boxes, which was possible because we used straight lines for the edges. Chaturvedi et al. [8] drew a bundled edge for edges between groups, resulting in the use of edge box overlap as a measure. As mentioned in Section 4.1, we utilized straight lines to represent links, and therefore, we could calculate the number of edge crossings, which simplifies the readability of a graph. We also calculated the edge-crossing angle which is known to affect readability [21]. However, this measure was identical for all four layouts; hence, we omitted it from the discussion presented below (54.1° for the maximum and 54.9° for the minimum). **Edge Crossings.** This metric counts the total number of edge crossings and should be small to improve readability. Notably, a long edge can overlap with many other edges along its path.

Uniform Edge Length. Uniform edge length is a measure representing the extent to which the lengths of edges in a graph are uniform; more-uniform edges make the graph easier to read. A survey paper by Gibson et al. explained this measure as an index for a regular structure that prevents the distortion of the graph [14]. We computed the variance of edge lengths in each graph as this measure, wherein a small variance implied high uniformity in the lengths of the edges in a graph, which is known to lead to high readability. The calculation of edge length variance was performed under the

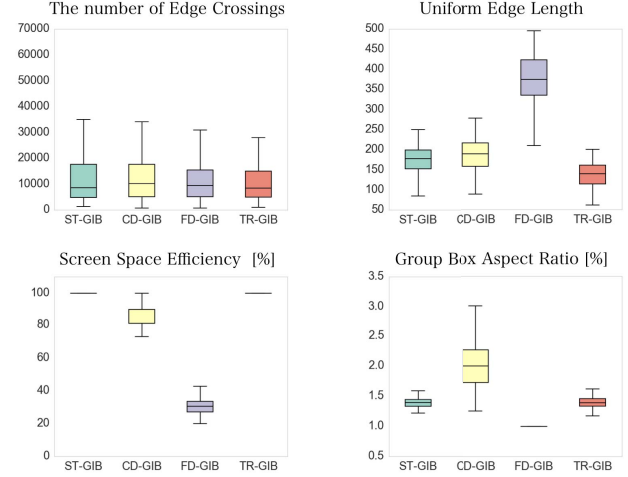


Figure 2: Computed measures of the tested graphs.

condition that the mean size of boxes was identical in all layouts.

Screen Space Efficiency. A two-dimensional screen-space-filling layout can be comprehensible [39]. In GIB, the efficient use of screen space allows each group to be drawn with more screen space. When screen space is limited, links can be too short to interpret, and the graph can become visually cluttered. For effective visualization, most of the screen space should be used to show information and this measure should be high.

Mean Group Box Aspect Ratio. Bruls et al. emphasized that boxes that are squarer in shape offer advantages including screen efficiency, easier comparison of box sizes, and improved accuracy of presentation [5]. They also observed that thin rectangular boxes can cause aliasing errors, whereas square boxes are easier to detect and point at. This factor may affect the result of the tasks described in Section 5.1 where participants are expected to estimate the area occupied by each box. Therefore, the aspect ratio should be close to unity to encourage readability.

Figure 2 summarizes the calculation results. When drawing the graphs, we minimized edge overlaps, aimed for uniform edge lengths, lower aspect ratios, and higher screen space efficiency. The results show that TR-GIB yielded the fewest edge crossings, and ST-GIB had the highest number of edge crossings with 541.2 as the mean number of links. In contrast, TR-GIB achieved the best result, and FD-GIB was the worst layout in terms of uniform edge length. Screen space efficiency was high for ST-GIB and TR-GIB, followed by CD-GIB. Furthermore, FD-GIB yielded the best aspect ratio, followed by ST-GIB and TR-GIB. ST-GIB and TR-GIB performed well in terms of both space efficiency and aspect ratio, whereas CD-GIB and FD-GIB performed well in terms of either space efficiency or aspect ratio. In summary, TR-GIB produced good results for all measures—especially for edge crossing and uniform edge length. CD-GIB and FD-GIB were poor in some measures; hence, we expect some trade-offs in the user study. ST-GIB produced average results, except for its highest number of edge crossings.

5 USER EXPERIMENTS

5.1 Tasks

We set up the tasks with reference to two previous studies: Vehlow et al. [40] and Saket et al. [38]. Vehlow et al. introduced four task taxonomies regarding the evaluation of clustered graph visualizations, namely group-only tasks (GOTs), group-vertex tasks (GVTs), group-edge tasks (GETs), and group-network tasks (GNTs). Of these four

Table 1: Parameters used for data generation

m_{mean}	m_{stdev}	m_{min}	m_{max}	V_{mean}	V_{stdev}
11.4	5.4	6	17	21.0	14.12
V_{min}	p_{in}	p_{group}	p_{bridge}	p_{out}	
4	0.0858	0.06	0.015	0.0006	

tasks, we used GOTs, GVTs, and GNTs for our GIB evaluation; GETs are appropriate for networks whose edges are grouped, thus, we do not use this taxonomy. Saket et al. report four types of tasks, namely GOTs, group-node tasks, group-link tasks, and GNTs, all of which could be applied, but we note that the GNTs used by Vehlow et al. are similar to the group-link tasks used by Saket et al. Vehlow et al. and Saket et al. also described examples of each task type; for instance, a group-node task would ask participants to identify the group with the most nodes. We selected the following tasks to test the readability of the four box-layout schemes.

- **Task 1 (Group Only Task).** How many groups are present in this graph?
- **Task 2 (Group-Node Task).** Which group has the maximum (minimum) number of nodes?
- **Task 3 (Group-Intra-Link Task).** Which group has the maximum (minimum) number of links?
- **Task 4 (Group-Inter-Link Task).** Which group has the maximum number of links connecting its nodes outside the group?

Task 1 is a GOT from Vehlow's taxonomy and is about the number of groups represented. It requires no node or edge information, but it does require aggregate group information. We expect this task to reveal which layout best helps users to understand the whole network structure.

Task 2 is from the GVTs and requires users to comprehend node information—specifically, the number of nodes. A GIB layout should be able to help users to easily estimate the size of a group, and this task should reveal which layout is the best at distinguishing the numbers of nodes within each group.

Task 3 is a GNT, which requires intra-link information. Intra-links connect a node to another node in the same group, whereas an inter-link connects a node to a node of another group. In a layout with good representation of intra-links, users can easily understand the relationships within each group, which is often critical in the analysis process. We expect this task to reveal the best layout for analyzing relationships within a group.

Task 4, another GNT, involves inter-link information. An inter-link, showing the relationship between a pair of groups, sometimes becomes valuable for representing the coupling between clusters in a graph similar to a social network. We suppose that this task can detect a good layout for this purpose.

In Task 2 and Task 3, the participants had to perform two subtasks: finding the maximum and minimum numbers of nodes or links. An effective GIB layout must display both large and small groups in a helpful manner. In other words, a layout is not practical when it is good at showing only groups with many or few nodes (links). For Task 2 and Task 3, we changed the task at the half of the total trial, from finding the maximum to finding the minimum. We did not set two subtasks in Task 4 because the group with the least connections to other groups was difficult to identify. The difficulty of this task would encourage participants to give random answers, which would contaminate our data.

We elicited two tasks from the GNTs, Task 3 and Task 4, because we aimed to assess the layouts' performance for representing both intra- and inter-links. As GIB is a method for visualizing the relationships within a group and among groups, the GIB layout method should be good at showing both intra- and inter-links.

5.2 Hypotheses

We constructed some hypotheses concerning our tasks. Based on a previous study [8], we know that the four GIB variants have different features. Thus, we expect the layout variants to affect the performance of the selected tasks. Each hypothesis has been constructed to focus on elements causing differences in the result.

- **Hypothesis 1.** In Task 1, participants performed best with the ST-GIB layout. Each box is organized in a row or a column in ST-GIB, which makes it easier for the user to simultaneously count more than one box.
- **Hypothesis 2.** In Task 2, ST-GIB yields the best results. ST-GIB places boxes in order of their sizes. Therefore, the biggest/smallest box, i.e., the correct answer of Task 2, is always set at the left-top/right-bottom of the screen. Participants should be able to easily detect the answer.
- **Hypothesis 3.** In Task 2, CD-GIB is not as accurate as the others. The aspect ratio is very rectangular with this layout, making it difficult to estimate the area at a glance.
- **Hypothesis 4.** In Task 3, TR-GIB yields the most accurate result. This layout performs as well as ST-GIB in terms of aspect ratio and screen space efficiency; hence, each group can occupy more area than that occupied using CD-GIB or FD-GIB. In addition, because we expect few edge crossings, the intra-edges can be seen with fewer interruptions caused by inter-edges.
- **Hypothesis 5.** In Task 3, FD-GIB does not produce better results than the others because its boxes are smaller than those in other layouts. If a box is small, the links within it are short, making it difficult to visually grasp the number of links.
- **Hypothesis 6.** In Task 4, TR-GIB and FD-GIB are more accurate than the other layouts. TR-GIB is good at avoiding edge overlaps, making it easy to recognize inter-links. FD-GIB wastes space but provides more area for inter-links. We expect that inter-links will be easy to see with these two layouts.

5.3 Experimental Design

We used a repeated-measures study design, and the independent variable of interest for each task was the GIB layout. Several values changed among the stimuli such as the number of groups in a graph, but they were normally distributed, and thus, we could ignore them.

In terms of the amount of test data, we generated 480 random networks in total; 120 were applied to each layout and were then allocated to the four tasks. Therefore, each task included stimuli with four layouts: 30 stimuli of the same layout in a task and 120 in total. Each task had six sub-blocks with 20 stimuli so that participants could have small breaks between the blocks.

In the experiment, each layout was tested on a distinct set of data to mitigate the effect of learning. As we corrected the gaze data of all participants for each network, we had to present the same set of networks to each participant in the experiment. With the repeated data for all layouts, participants would see each network four times with different layouts, which may result in some learning effect.

Participants may have become confused if we changed the task in every trial; hence, the participants continuously performed the same task continuously until the next break. We randomized the stimuli order for each task such the layout shown to participants changed during each trial. We also randomized the task order to mitigate the effects of learning and fatigue. Participants took brief breaks of roughly 30 s after every 20 stimuli, a sub-block, and a relatively long break after one task, which involved 120 stimuli. We empirically set the color as gray, and the width of edges were set as shown in Figure 1.

5.4 Study Procedure

Before the experimental sessions, participants were asked to fill out a questionnaire about their age, gender, and field of study. Next, they read a manual regarding biometric measuring, eye tracking, and each GIB layout. We then explained the experiments in detail, followed by a tutorial to check whether the participants could interpret the



Figure 3: Screenshot of the stimulus in Task 2, in which the participants identified the largest box.

network and solve our tasks. The trial data in the tutorial differed from the data in the experiments but provided participants with a practical guide to the experimental procedure.

Once the participants completed all of the abovementioned steps, they proceeded to the actual experiments. They worked on the four tasks in random order to compensate for the leaning and fatigue effects, and 120 trials were included in each task. For Tasks 2 and 3, we asked the participants two questions about the maximum or minimum; the first 60 trials asked for them to identify the maximum, and the last 60 trials asked for the minimum, wherein we permuted the trials such that each half presented 15 networks for each layout during each task.

There was no time limit on the tasks, and the answers were recorded on a server. The participants were instructed to focus on answering correctly rather than quickly. If participants focused on answering quickly, we were likely to encounter higher error rates and more chaotic gaze trajectories, which was not in accordance with the study's aims.

Figure 3 shows a screenshot of the actual stimulus (Task 2). Participants clicked on the biggest box. Participants could change their selection by clicking on another box. When they pressed the enter key, the answer and the completion time was sent to the server; then, the next stimulus was shown.

We conducted this eye-tracking experiment in a room with artificial illumination and few objects. The participants' mobile phones were switched off to reduce distractions during the experiments. The eye movements were recorded using a Tobii Pro X3-120 eye-tracking system with a screen resolution of $1,920 \times 1,080$ pixels. The participants sat in front of a display at a distance of 65 cm. In the software of the eye tracker, we used an I-VT filter to determine whether the gaze was a fixation or a saccade, as described in [31].

5.5 Participants

We used a within-subject study design with 20 participants (8 women and 12 men). Technical difficulties affected the data collected from one participant; these data were excluded from the study. Thus, we analyzed data from 19 participants in total. Their average age was 20.8 years, with the youngest participant being 18 years old and the oldest participant being 24 years old. All participants were Japanese students at Kyoto University; four participants were studying engineering, and the others had different majors. Notably, Japanese is read from left-to-right, which might affect the task performance as described in [41]. All participants had normal or corrected-to-normal color vision. Each participant was compensated with 3,000 Yen. The experiment duration was 1.5–2 h, including the explanation and breaks.

6 EXPERIMENTAL RESULTS

We recorded the task accuracy, completion time, and gaze data during the user experiments. The result of the experiments are

described below in two groups: the accuracy and completion time is described in Section 6.1; eye-tracking data based on our hypotheses is described in Section 6.2.

6.1 Task Results

In this subsection, we describe the task-performance results of accuracy and completion time. As the data were not normally distributed, we used non-parametric analysis. We conducted the Friedman test to determine whether the data was significantly different, followed by the post-hoc Wilcoxon signed-rank test. In all cases, we used $p = 0.05$ as the significance threshold. Hereafter, the statistical analyses for each task are discussed separately. We assume that each layout offers advantages and disadvantages and that all tasks were independent.

Task 1. The mean accuracy for all layouts was 97.7%, and the mean completion time was 4480 ms. The Friedman analysis showed significant differences in the completion time ($p < 0.001$), whereas no significant differences were found in task accuracy. The post-hoc Wilcoxon test showed that participants completed this task significantly faster with the ST-GIB and TR-GIB layouts than the CD-GIB and FD-GIB ($p < 0.001$) layouts; in addition, CD-GIB was faster than FD-GIB ($p < 0.001$).

The time required for ST-GIB in this task was approximately 500 ms less than that required to complete the task with the CD-GIB and TR-GIB layouts and 900 ms shorter than that required with FD-GIB. These results support Hypothesis 1.

Task 2. ST-GIB was the best layout in terms of accuracy, with a mean accuracy of 89.6% and completion time of 2526 ms. FD-GIB provided the shortest completion time, with a mean accuracy of 83.0% and completion time of 2425 ms. According to the Friedman test, both correctness ($p < 0.001$) and completion time ($p = 0.0025$) for this task were significantly different among the four layouts. The Wilcoxon test showed that ST-GIB was significantly more effective than all others ($p < 0.001$ for CD-GIB and FD-GIB; $p = 0.0019$ for TR-GIB), and CD-GIB was relatively ineffective ($p < 0.001$ for ST-GIB; $p = 0.0041$ for FD-GIB; $p = 0.0019$ for TR-GIB).

These results support Hypothesis 2 and Hypothesis 3. FD-GIB and TR-GIB performed better than CD-GIB, but they were not as effective as ST-GIB, which places boxes in order of size.

Task 3. FD-GIB returned the highest accuracy (78.8%) and the longest completion time (3383 ms). Based on the Friedman test, both correctness and completion time ($p < 0.001$) were significantly different among the four layouts. According to the Wilcoxon test, FD-GIB performed better than all other layouts in terms of correctness ($p < 0.001$ for ST-GIB and CD-GIB; $p = 0.017$ for TR-GIB) and completion time ($p = 0.007$ for ST-GIB; $p < 0.001$ for CD-GIB and TR-GIB). TR-GIB returned significantly more correct responses than ST-GIB ($p = 0.039$) and CD-GIB ($p = 0.049$), and participants finished Task 3 more quickly with ST-GIB than they did with CD-GIB ($p = 0.014$) and TR-GIB ($p = 0.014$).

These results contradict Hypothesis 4 and Hypothesis 5; FD-GIB allowed participants to complete this task with the best accuracy and shortest completion time. TR-GIB did not produce bad results, but its accuracy was 6% lower and participants needed extra 500 ms to complete this task as compared with FD-GIB.

Task 4. The mean accuracy of all layouts was 61.2% and the mean completion time was 5405 ms. The Friedman test shows that only the completion time ($p = 0.039$) was significantly different among the four layouts. The Wilcoxon test shows that participants reached their answers 400 ms faster with TR-GIB than with FD-GIB ($p = 0.003$); hence, Hypothesis 6 was disconfirmed.

6.2 Eye-tracking Results

This section presents the eye-tracking results and discusses how they relate to the hypotheses presented above. Eye-tracking data can be analyzed with quantitative measures, such as fixation duration or

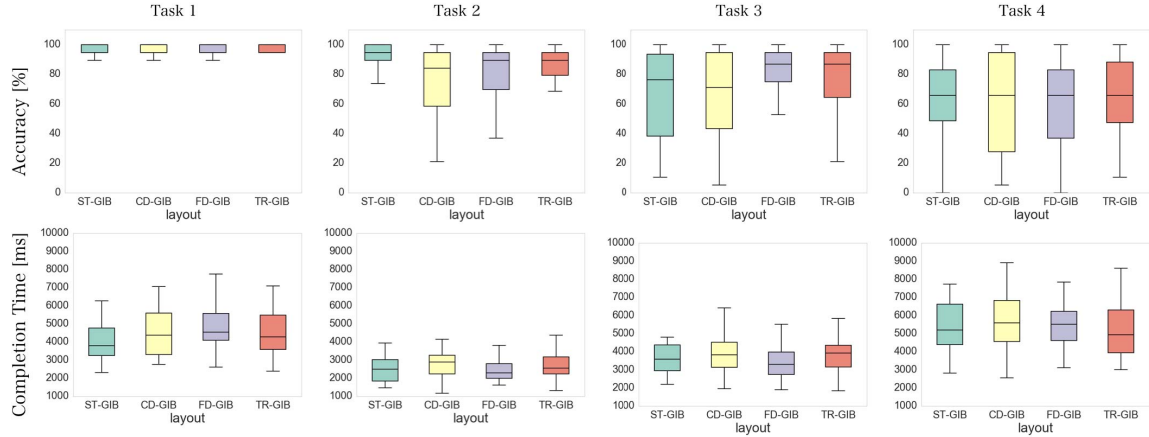


Figure 4: Task-performance results. The upper row represents the accuracy for each task, and the lower row lists the completion time.

saccade length, or with qualitative measures such as visualizations of the eye-gaze trajectories. As a quantitative measure, we counted the number of distractors that a participant looked at before or after the participant first looked at the correct answer to the task. Any box other than the correct answer is considered a distractor. The number of boxes a participant looked at before first seeing the correct answer is **DB**, and **DA** represents the number of boxes a participant looked at after seeing the correct answer.

Qualitative analysis uses visualizations of the eye-tracking data such as a trajectory map or a gaze heat map. We followed Burch et al.'s example when analyzing the eye-tracking data [6]. Figure 5 shows a time-varying visualization based on AOIs in the task stimuli. For each combination of a task and a layout, the left figure represents the color-coded AOIs that each participant looked at over the duration of the task, whereas the right figure shows time-series plots that show how many participants looked at specific AOIs in each moment of the task. Participants in the left color-coded figure are lined up in order of accuracy, and they are also ordered according to the completion time; thus, the top participant is the one who solved the task correctly with the shortest completion time. For the data model and transactions, we aligned all trajectories to common start and end times, and the completion times were normalized and linearly mapped to a scale of 1,000 units. We observed that, separately, the color-coded bars look very similar for each layout; fast and slow participants had similar fixation patterns, especially in Task 2 and Task 3. This indicates that participants use a similar strategy for each task, regardless of the task completion time. Note that the main purpose of the eye-tracking data is to uncover the elements of a GIB layout that help or hinder task performance. Therefore, the following discussion focuses on how the eye-tracking data relates to our hypotheses about visual attributes in GIB.

Hypothesis 1. This hypothesis was confirmed above as ST-GIB offered the highest task accuracy. To verify the hypothesis, we computed the gaze count as listed in Table 2. ST-GIB did not have as many gaze counts as the other layouts, which indicates that

participants recounted the boxes less, or participants were able to count the number of groups without moving their gaze. Figure 5 shows that each participant looked at AOIs in almost the same order in case of the ST-GIB and TR-GIB layouts, which indicates that all participants used similar strategies for the task because these layouts had less visual clutter. The time-series plot indicates that the strategy was counting the number of boxes from the left top of the screen, perhaps Japanese participants usually read from left-to-right and top-to-bottom. We suppose the well-aligned boxes in ST-GIB encouraged fewer gaze counts and this robust strategy.

Hypothesis 2. ST-GIB achieved the best results for Task 2, which supports this hypothesis. As shown in Table 2, ST-GIB and FD-GIB did not require participants to gaze at as many distinct regions as compared with CD-GIB and TR-GIB. Another measure, **DB** supports this hypothesis and is shown in Table 3, where the **DB** of ST-GIB was smaller than the others. This indicates that participants could easily reach the correct answer with the ST-GIB layout, likely because the boxes were sorted by size.

Figure 5 (b) also supports this hypothesis as the boxes that participants looked at just before completing the area comparison were placed close to each other. We conclude that participants could easily compare the size of the candidate boxes with the ST-GIB layout.

Hypothesis 3. This hypothesis is supported by the results of Task 2, wherein CD-GIB performed the worst in terms of accuracy. We suppose that this deterioration in performance is due to the diverse aspect ratios of the boxes, which might have confused participants. We also found a clue to reveal which element in this layout interrupted the task performance in Figure 5 (b), wherein two boxes with many fixations are placed apart. In the CD-GIB layout, boxes with similar areas are sometimes placed far apart; hence, the areas might be difficult to compare.

TR-GIB had many more gaze counts than CD-GIB, as shown in Table 2. Furthermore, boxes that are candidates for area comparison are often located close to each other because the TR-GIB is generated by reordering the tiles in ST-GIB while maintaining the structure of the treemap. This arrangement makes the comparison relatively easy.

Table 2: Mean gaze counts (standard deviation).

	ST-GIB	CD-GIB	FD-GIB	TR-GIB
Task 1	12.5 (5.49)	14.4 (6.79)	15.6 (6.85)	14.1 (6.31)
Task 2	8.6 (5.11)	9.2 (5.30)	8.3 (4.04)	9.1 (5.18)

Table 3: Gaze count of distractors in Task 2 (standard deviation).

	ST-GIB	CD-GIB	FD-GIB	TR-GIB
DB	3.6 (3.59)	3.9 (3.67)	4.1 (3.48)	3.7 (3.06)
DA	2.84 (3.33)	2.99 (3.67)	4.1 (3.48)	3.7 (3.06)



Figure 5: AOI-based visualization for the time-varying analysis of eye-tracking data. The left figure in each row and each column represents the color-coded time-varying AOIs that were focused on by each participant. The right figure shows time-series plots that express the relative number of participants who visited specific AOIs. The participants are ordered according to task accuracy and completion time, and the bars of the three AOIs closest to the answer are larger than the others in the left figure.

Therefore, we suppose that the poor aspect ratio and the separation between boxes with similar areas resulted in low task accuracy.

Hypothesis 4. TR-GIB did not produce the best results in Task 3. Hence, this hypothesis was not confirmed, although the layout resulted in significantly higher task accuracy than ST-GIB and CD-GIB. Figure 5 (c) shows that the main strategy in this task was to begin by identifying large or small boxes. After the large boxes were recognized, participants compared several boxes to decide which group had more intra-links. As ST-GIB and TR-GIB achieved good results in Task 2, they might have helped participants with the first process. This is also supported by the finding that these layouts had the lowest DB in this task as shown in Table 4. In the second process, participants tended to fixate on boxes with many intra-links in TR-GIB, whereas the fixations were dispersed in ST-GIB, as shown in Figure 5 (c). We suppose that TR-GIB was effective in the second process because of the small number of edge crossings. As intra-links are presented more clearly with fewer edge crossings, TR-GIB also achieved good results in the second process.



Figure 6: Trajectory map for Task 4. The gaze movement is represented with a pink line and gaze aggregation can be seen on the bundles of inter-links.

Hypothesis 5. As FD-GIB achieved the best result in Task 3 despite its small boxes, this hypothesis was refuted. As FD-GIB was not helpful for in Task 2, this layout might have been unhelpful in the first process of the strategy adopted in Task 3, as mentioned above. The DB of this layout was relatively large, which also signifies the difficulty of this process. Therefore, the superiority of FD-GIB might indicate its helpfulness in the second process. Table 4 shows that the DA of FD-GIB was the smallest. This result indicates that the participants cannot easily estimate the number of intra-links visually ambiguously; we suspect that the coloration of the links explains this trend. As we drew the edges in grayscale, the number of edges could be seen as dark gray shading, representing the degree of aggregation. Moreover, the boxes in FD-GIB are much smaller than those in the other layouts. Thus, this trend might be particularly remarkable as the links are more dense within each box.

Hypothesis 6. We found no significant difference in the accuracy of Task 4. Hence, this hypothesis could not be confirmed. The four layouts have differences in edge crossings and the uniformity of the edge lengths, which are thought to affect the readability of a graph—especially link information—but these factors did not yield any significant differences. The completion time of TR-GIB tended to be shorter than that of FD-GIB.

Figure 5 (d) shows that participants' fixations were dispersed across many AOIs, except when the network was laid out with CD-GIB. We found that the participants' gaze was concentrated at the center box in CD-GIB regardless of whether it was the correct answer. The center box is connected to the largest number of boxes with this layout; hence, the inter-links are often drawn near this box. These sparse and centered fixations indicate that participants searched for the answer by focusing on bundles of inter-links. Because certain pairs of groups had strong relationships in our data, the bundles of inter-links in a graph were key to finding the answer to this task. Figure 6 shows a trajectory map of the gaze data for this task, and the fixations are concentrated along the link bundles. When we search for a link bundle, the longer bundles that are undesirable for graph drawing often stand out more than shorter bundles. The long bundle of inter-links might have affected the result in this task, obscuring possible differences in task accuracy.

Table 4: Gaze count of distractors in Task 3 (standard deviation).

	ST-GIB	CD-GIB	FD-GIB	TR-GIB
DB	3.8 (4.64)	5.1 (5.11)	4.8 (4.38)	3.8 (4.48)
DA	4.8 (4.65)	4.8 (4.84)	3.7 (4.27)	5.3 (5.37)

7 DISCUSSION

Our eye tracking experiment helped us to find some interesting results about GIB. We describe them regarding tasks in this section.

Task 1. This task was very simple, but the completion times were significantly different for the four layouts. Figure 5 (a) suggests why the well-aligned ST-GIB was effective in this task. We also found that FD-GIB required the longest time to detect the answer as the boxes were scattered. We suppose that well-arranged boxes led participants to use robust strategies to count the groups from the left top of the screen.

Task 2. In this task, ST-GIB achieved the best accuracy, wherein boxes were arranged in order of area, and the aspect ratio of each box was low and uniform. CD-GIB, with boxes of high aspect ratios, returned lower accuracy than the other layouts, whereas FD-GIB was the most accurate and the fastest layout; FD-GIB had boxes with the lowest aspect ratios. Therefore, a layout with low aspect ratios is helpful when identifying the largest or smallest box.

We also found that the distance between boxes of similar sizes is important. Figure 5 (b) shows that these boxes can be placed far apart in the CD-GIB layout. When these boxes are placed next to each other, as in ST-GIB and TR-GIB, the size comparison is easier.

Task 3. Participants were required to detect the box with the most or fewest intra-links. For Hypothesis 4, we assumed that the number of edge crossings was important in this task, which would indicate that intra-links are difficult to distinguish from inter-links. TR-GIB and FD-GIB, which are the layouts with the fewest edge crossings, returned higher accuracies than the others. The importance of edge crossings has been frequently discussed in the literature, and this result confirms the importance of this factor.

Quantitative measures of eye tracking, namely **DB** and **DA**, and Figure 5 (c) also support this result. Participants searched for the large/small boxes to solve this task in the first step, and TR-GIB was good in terms of comparing box sizes, which might make it easier to obtain the correct answer. The fewest number of edge crossings and the most uniform edge lengths also helped in obtaining an accurate answer. FD-GIB had the highest accuracy and the smallest **DA**. This indicates that the small boxes in FD-GIB were large enough to complete this task and encouraged participants to find the correct answer quickly. The smallest **DA** also means that little comparison was needed after fixating on the correct answer for the first time. In addition to this result, we suspect that participants could solve this task by seeing only the color of a box, which represents the density of links. This result indicates that the small boxes of FD-GIB do not interrupt performance on this task.

We chose this task to reveal which layout is the best for representing intra-links. Based on a survey about graph drawing for networks with group structure [38, 40], another task is suitable for this purpose—finding the vertex of the highest degree within a particular group. This task requires more concrete data, and FD-GIB might be not appropriate for this task owing to its small boxes and heterogeneous edge lengths. We revealed the effectiveness of FD-GIB and TR-GIB in this task, but some computation measure showed some differences between them; hence, testing participants with another task would be interesting.

Task 4. We did not find any differences in task accuracy, and participants did not use as much time to complete this task with TR-GIB as they did with FD-GIB. This can be attributed to various factors. However, in this study, we were unable to find a difference in the visibility of inter-links.

Summary. Each layout had different trade-offs depending on the type of user task. The differences of the mean values within a task were sometimes small, but we found that the variances or minimum values tended to considerably differ. For example, the minimum correctness of FD-GIB was much higher than that of TR-GIB in Task 3, although the mean accuracy varied by only 6%. ST-GIB achieved the best results on Task 1 and Task 2, whereas CD-GIB

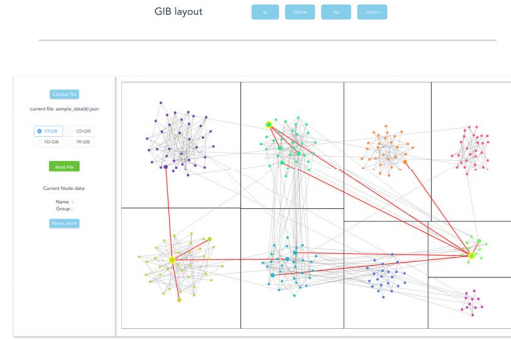


Figure 7: The website we constructed. This site provides the four GIB layouts together with further interactive analysis.

did not achieve a good result. FD-GIB performed well in all tasks, and it provided the best results in Task 3. Participants also achieved averages scores on all tasks in TR-GIB.

For practical analysis of real-world data, a graph must comprehensively display the links. In Task 3, FD-GIB and TR-GIB achieved good results and were average in other tasks; thus, we conclude that these two layouts are the best of the four. As we found some trade-offs in the task results, we recommend selecting a layout based on the purpose of a given analysis. If a user wishes to grasp aggregate and abstract information about the graph, we suppose that FD-GIB is the best. Participants completed Task 2 and Task 3 fastest with the FD-GIB layout, wherein a box is represented as small; thus, this layout is suited for quick and uncomplicated tasks. In contrast, TR-GIB displays link information more concretely owing to its efficient use of screen space. This layout did not result in bad scores in any tasks; hence, it may be more useful for clearly displaying specific information about the links in a network.

8 WEBSITE FOR THE GIB LAYOUTS

As a part of our work, we have created an open-access website for the readers to easily access the four GIB layouts [2]. Users can easily prepare GIB layouts as bitmaps or svg image files using any set of network data. Figure 7 shows a screenshot of the website. Nodes can be selected with a click of the mouse, and then the node will be highlighted along with the links connected to it.

9 CONCLUSION

Herein, we evaluated four GIB layouts through user tests while collecting eye tracking data. We determined which layout is the best for completing four tasks, and the eye tracking data supports the conclusions thus obtained. The best layouts were FD-GIB and TR-GIB; both layouts offered different advantages and disadvantages. FD-GIB performed well when determining the number of links, abstract information, whereas TR-GIB proved better for representing more concrete relationships, such as the links between specific nodes. ST-GIB performed well in terms of understanding the box size; however, participants could not interpret links well with this layout.

The eye-tracking data provided evidence to support the results as well as interesting clues that reveal which elements in a visualization affect task performance. For example, the clear alignment of ST-GIB helped participants to count the number of groups. We also found that the candidates for comparison should be placed next to each other. Furthermore, the inefficient use of screen space in the FD-GIB layout helped participants to estimate the number of intra-links. These elements can be used to predict the readabilities of other graph-drawing methods.

We have also created an open-access website through which anyone can easily access the four GIB layouts. We hope to encourage other researchers to have new findings using this website.

The present work does have a few limitations. Other methods could be used to generate random network data. We used only one set of network parameters in the user tests; thus, other cases may be of interest. Additionally, nodes and edges may be expressed and arranged in several other ways, and these choices may also influence the results. Testing another task may also reveal further results or larger differences than those observed by us.

The massive eye-tracking dataset can be further analyzed in future work. For example, the difference between the performances on different stimuli of individual participants could be analyzed. We also plan to evaluate other visualization methods for graphs with group structure to compare their results with the results of this study.

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